

PROJECT PEAR

Quantitative Valuation & Momentum Analysis

A Major Consumer Technology Company

Analysis Date: 2026-06-18

Data Period: 2021-06-22 to 2026-06-18

Key Findings



Valuation

DCF median fair value of \$101 implies -66.1% return. The 90% CI is [\$72, \$143].
Trading at a premium to all peers. P/E z-score of 1.28, P/B z-score of 1.73.
The gap between DCF and market price reflects optionality, quality premium, and buyback yield not captured by the model.

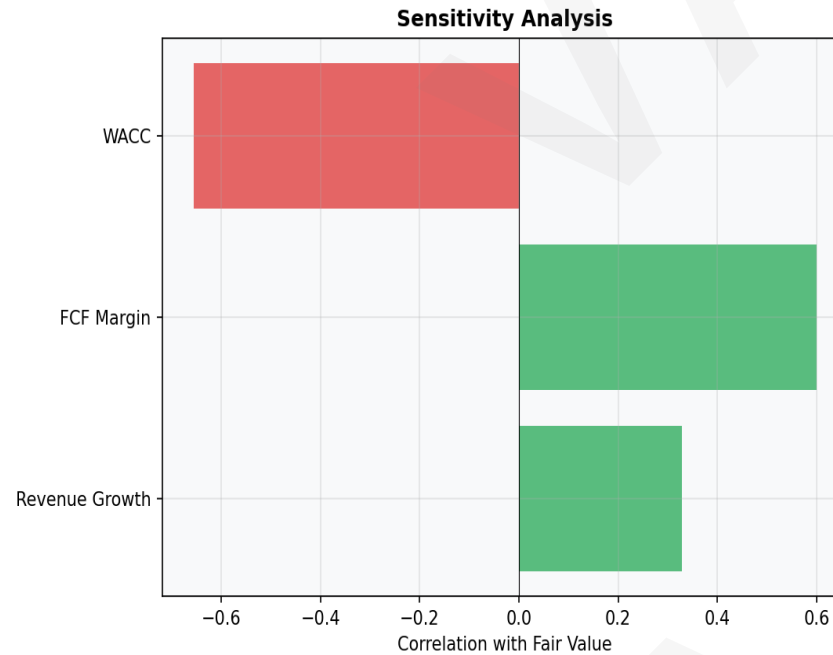
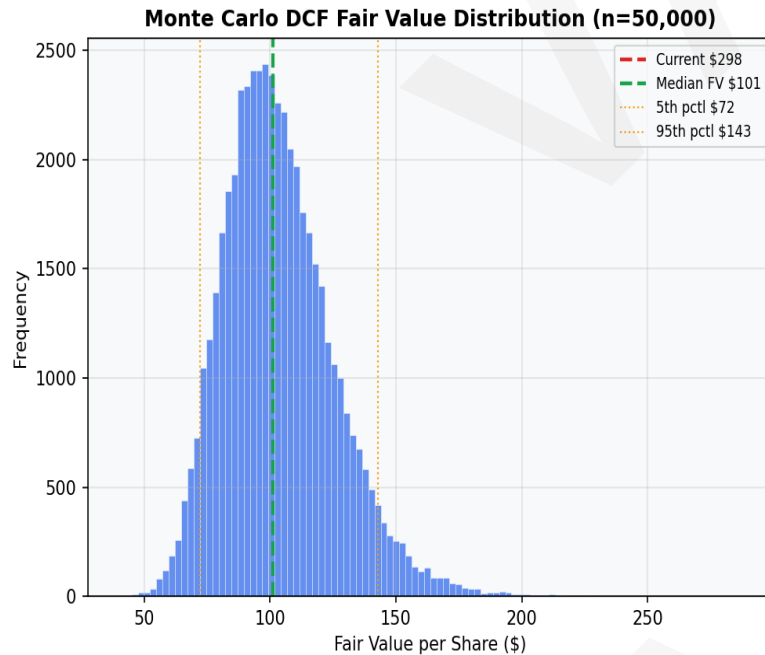
Momentum

Long-term bullish (SMA50 > SMA200, 1yr return 52.8%). Short-term neutral (RSI 39, MACD bearish).
Hurst exponent 0.446 suggests mild mean-reversion. Bootstrap CI [0.368, 0.549] spans 0.5.
63-day relative strength vs S&P 500 at 1.054 shows modest outperformance.

Events

2 statistically significant events. Buyback announcement +8.6% CAR (p=0.028). Q1 FY2024 earnings -10.0% CAR (p=0.005).

Monte Carlo DCF Valuation



Model Parameters

Simulations: 50,000

Beta: 1.086

WACC: 10.27% (base)

FCF Margin: 23.7%

Growth: 2%/6%/12% (tri)

Terminal Growth: 2.5%

Output

Median FV: \$100.97

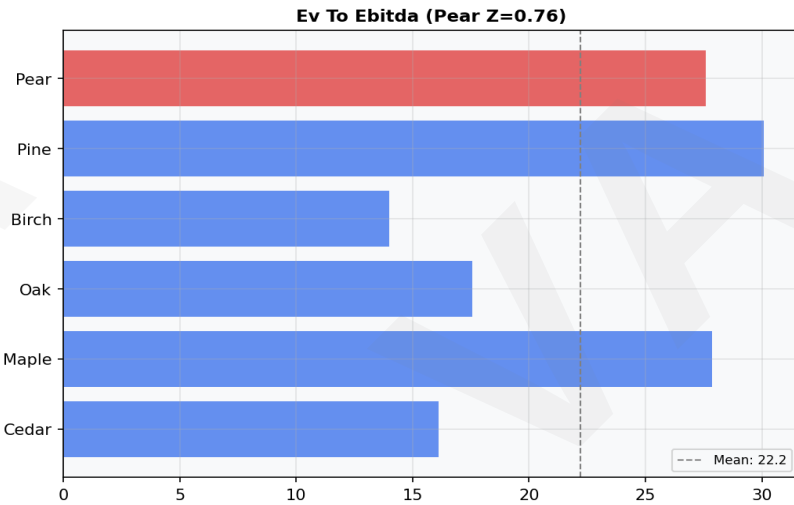
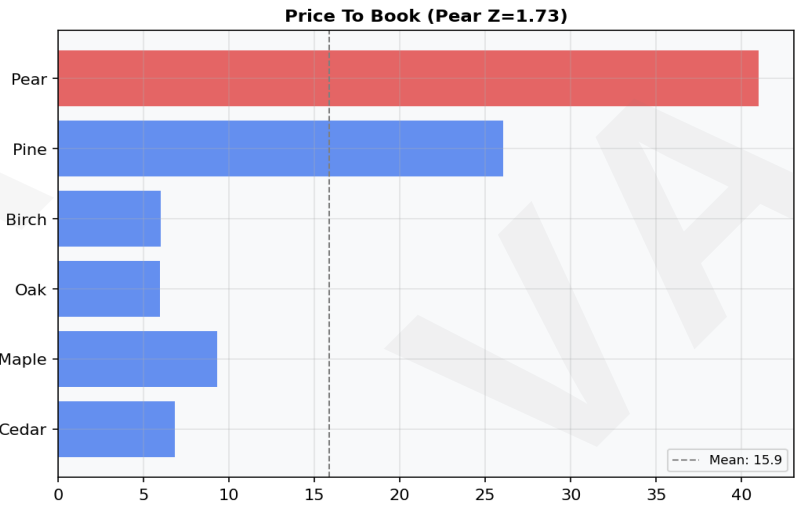
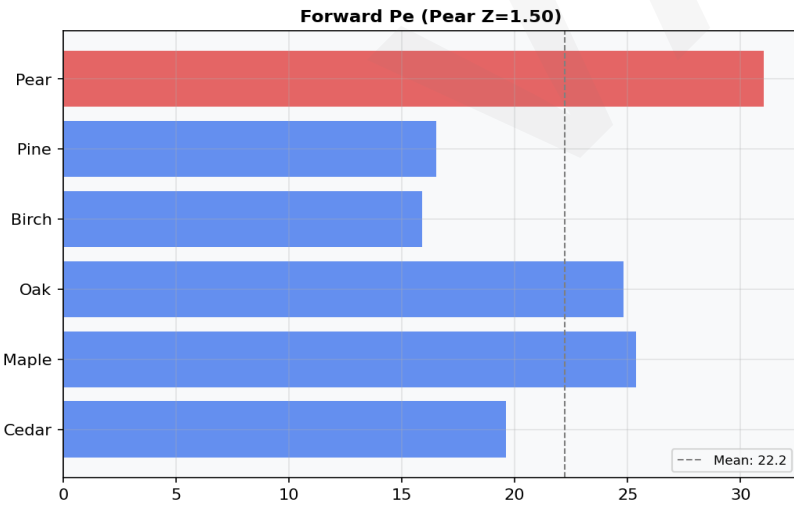
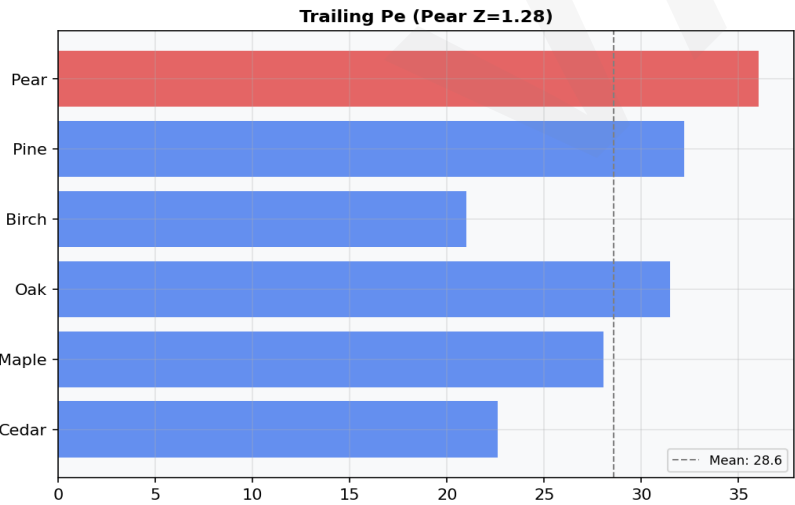
90% CI: [\$72, \$143]

P(Undervalued): 0.0%

The DCF cannot justify the current price on standard assumptions. This is expected for a company trading at 36x earnings. The model does not price optionality in new product categories, services ecosystem lock-in, or the buyback programme's mechanical support. An investor should read this as 'the base business is priced beyond its cash flows' rather than 'the stock will fall 66%'.

Relative Valuation vs Peers

Pear Corporation - Relative Valuation vs Peers



Z-Scores

Trailing Pe: Z=1.28

Forward Pe: Z=1.50

Price To Book: Z=1.73

Ev To Ebitda: Z=0.76

Pear trades at a premium on every metric. The P/B premium is extreme ($z=1.73$) but reflects a capital-light model and decade of buybacks. EV/EBITDA ($z=0.76$) is the most moderate premium, a fairer measure for cross-company comparison.

Momentum Analysis

Pear Corporation - Price Action & Technical Indicators



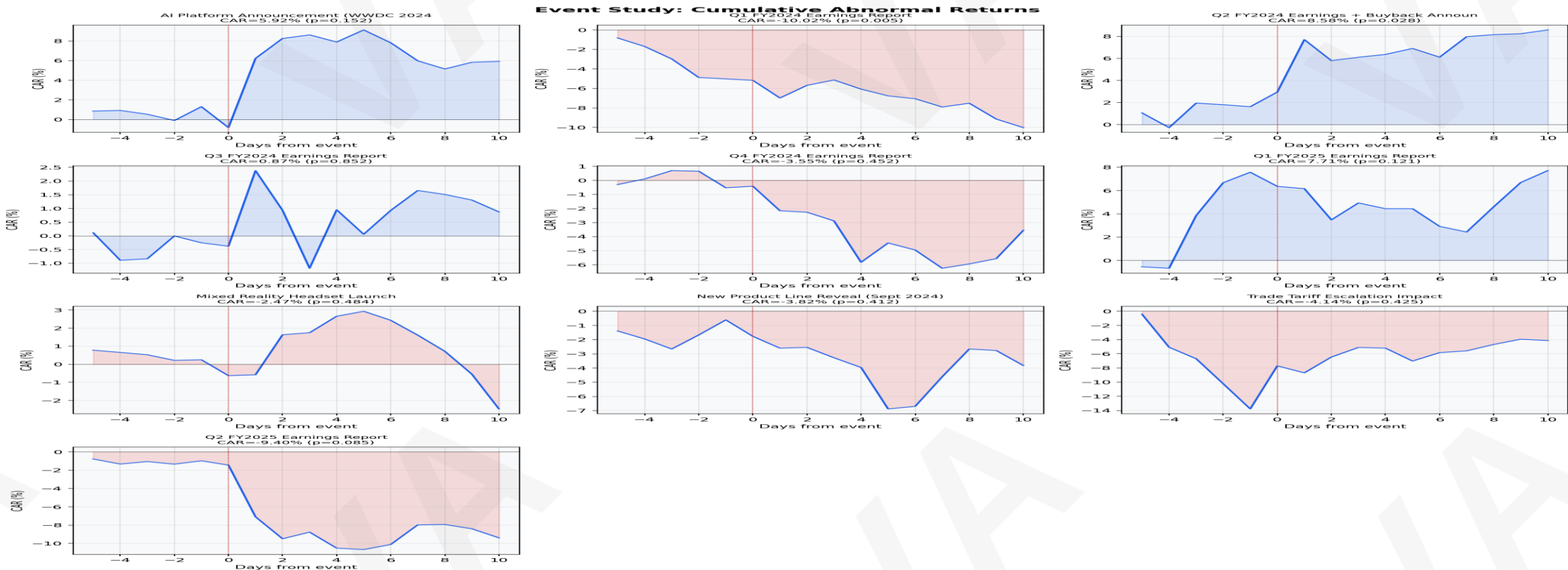
Current Signals

Trend	BULLISH
RSI (14)	39 (Neutral)
MACD	BEARISH
Bollinger %B	0.32
Hurst	0.446 (Mean-Rev)

Momentum Returns

1 Week	0.72%
1 Month	-0.40%
3 Months	19.72%
6 Months	8.64%
1 Year	52.81%

Event Study Results



Two statistically significant events at 5% level. The buyback announcement (May 2024) generated the largest positive CAR (+8.58%, $p=0.028$), suggesting the market values capital return commitments strongly. The Q1 FY2024 earnings miss drove the largest negative CAR (-10.02%, $p=0.005$). The AI platform announcement at WWDC 2024 was positive (+5.92%) but not significant ($p=0.15$). The trade tariff escalation (April 2025) and Q2 FY2025 earnings both show negative CARs but fall short of significance.

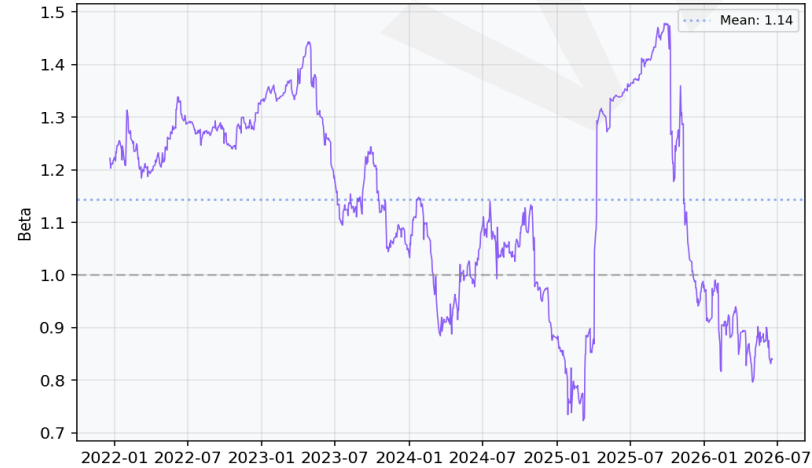
Volatility & Risk Profile

Pear Corporation - Volatility & Risk Profile

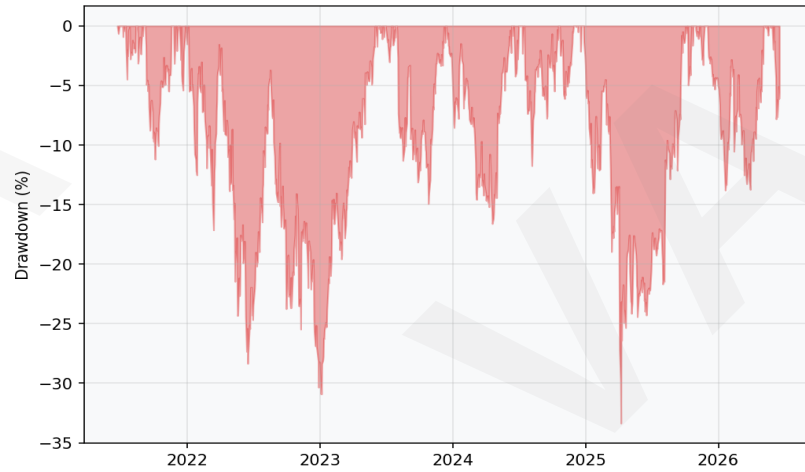
GARCH(1,1)-t Conditional Volatility



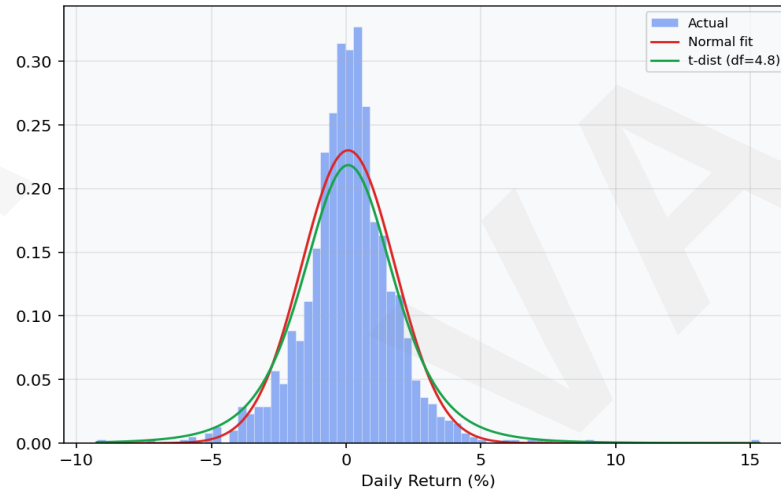
Rolling 6-Month Beta vs S&P 500



Drawdown Profile



Return Distribution



Risk Metrics

VaR 99%: -4.45%

CVaR 99%: -5.60%

Max Drawdown: -33.4%

Sharpe: 0.566

Sortino: 0.826

Vol Regime: MODERATE

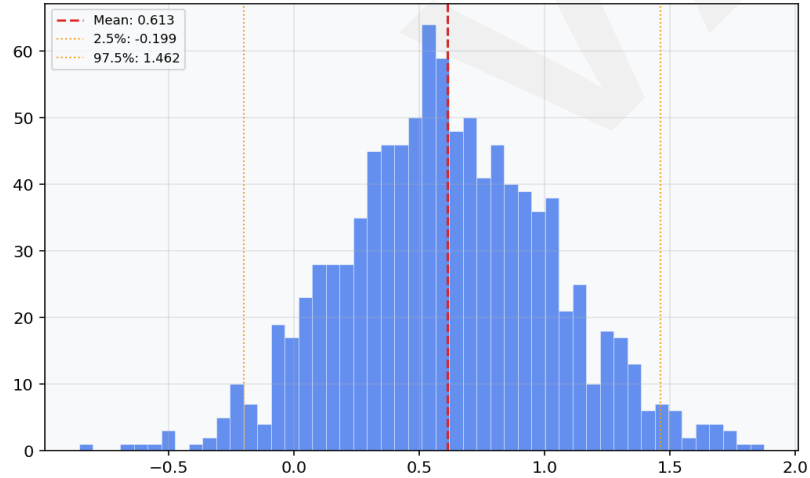
Current Beta: 0.839

GARCH persistence: 0.9762

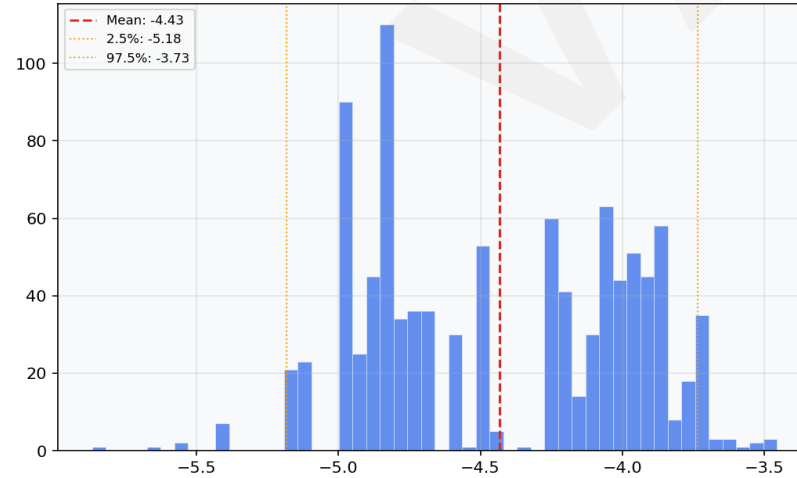
Statistical Robustness

Bootstrap Confidence Intervals (1,000 Block Bootstrap Iterations)

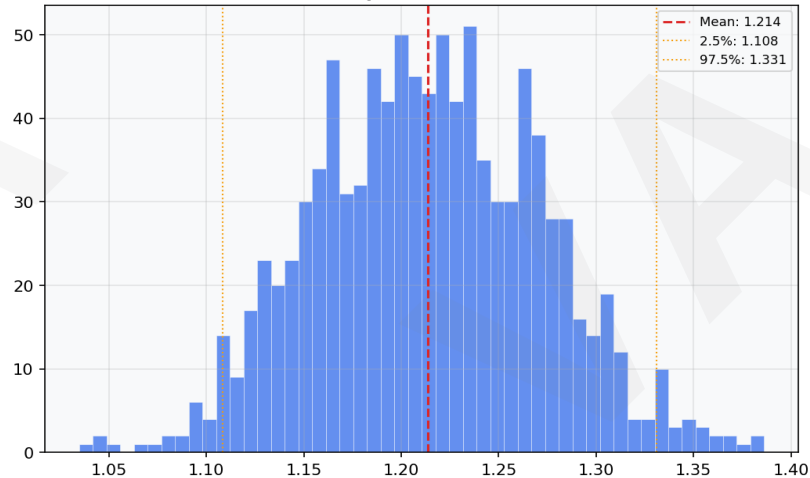
Bootstrap Distribution: Sharpe Ratio



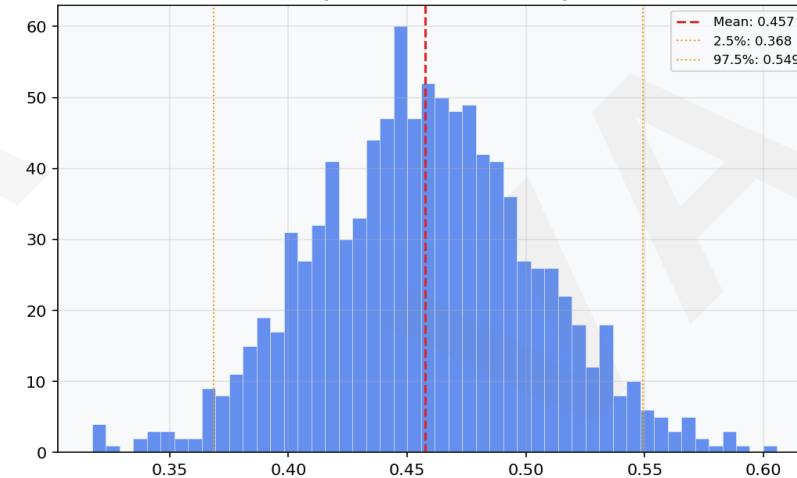
Bootstrap Distribution: VaR 99% (Daily %)



Bootstrap Distribution: Beta



Bootstrap Distribution: Hurst Exponent



Bootstrap 95% CIs

Sharpe: [-0.199, 1.462]

VaR 99%: [-5.18%, -3.73%]

Beta: [1.108, 1.331]

Hurst: [0.368, 0.549]

Sharpe CI includes zero. We cannot statistically confirm positive risk-adjusted returns. Hurst CI spans 0.5, so the mean-reversion finding is not conclusive. Beta CI is the tightest, indicating reasonable stability in systematic risk. Block bootstrap (size=20) preserves autocorrelation structure.

Known Limitations

DCF does not price optionality in new business lines, AI monetisation, or services ecosystem growth

Terminal value dominates the DCF output. Small parameter changes produce large valuation swings

Buyback programme reduces shares over time. Static share count understates per-share fair value

Technical indicators are backward-looking. Predictive power is weak in efficient large-cap markets

Event studies use single-factor market model. Multi-factor models would isolate abnormal returns better

Beta instability (range 0.72 to 1.48) makes any single-point CAPM-based WACC questionable

Daily close prices miss intraday dynamics, options signals, and after-hours reactions

All models assume past is informative about the future. Structural changes could invalidate parameters

Conclusions

1. DCF says overvalued. Market says otherwise.

At 36x earnings, a standard 5-year DCF cannot justify the price. The 66% implied downside reflects what the model cannot capture (optionality, quality premium, buyback mechanics). An investor must decide whether those premiums are justified.

2. Momentum is positive but fading.

Long-term uptrend intact. Short-term indicators cooling (MACD bearish, RSI neutral, 1-month momentum flat). Consistent with consolidation, not reversal. Hurst exponent suggests mild mean-reversion weakens pure momentum case.

3. Events matter more than earnings beats.

92% earnings beat rate shows strong execution. But the buyback announcement drove the largest positive reaction. Forward guidance and macro shocks (tariffs) matter more than marginal earnings surprises at this valuation.

What to watch.

Services revenue growth rate (justifies the optionality premium if it accelerates)

AI product adoption metrics (could shift the growth trajectory)

Capital allocation (continued buybacks support price mechanically)

Trade policy developments (tariffs have driven the largest recent drawdown)